Geometry

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Area and perimeter**

A rectangle has the dimensions shown below. Find its area and perimeter.

Area $= 12 \times 7 = 84$ square units. Perimeter $= 2(12 + 7) = 38$ units.

2 **[Foundational]** • **Target: Area and perimeter**

Find the area of a trapezoid with parallel sides 8 and 14, and height 6.

Area $= \frac{1}{2}(8 + 14)(6) = \frac{1}{2}(22)(6) = 66$.

3 **[Foundational]** • **Target: Area and perimeter**

A triangle has base 15 and height 10. Find its area.

Area $= \frac{1}{2}(15)(10) = 75$.

4 **[Foundational]** • **Target: The Pythagorean theorem**

A right triangle has the two legs shown below. Find the length of the hypotenuse c.

$c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15$.

5 **[Foundational]** • **Target: The Pythagorean theorem**

A 25-foot ladder leans against a wall, with its base 7 feet from the wall. How high up the wall does the ladder reach?

$h = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = 24$ feet.

6 **[Foundational]** • **Target: Volume and surface area**

Find the volume of a cylinder with radius 4 and height 9, in terms of π.

$V = \pi r^2 h = \pi(16)(9) = 144\pi$.

7 **[Foundational]** • **Target: Volume and surface area**

Find the surface area of a rectangular prism with dimensions 5 by 4 by 3.

$SA = 2(lw + lh + wh) = 2(20 + 15 + 12) = 2(47) = 94$ square units.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Similar triangles and proportional reasoning

Triangle ABC is similar to triangle DEF, with $AB = 8$, $DE = 12$, and $BC = 10$. Find EF.

Scale factor $= \frac{DE}{AB} = \frac{12}{8} = 1.5$. $EF = 10 \times 1.5 = 15$.

9 [Intermediate] • Target: Similar triangles and proportional reasoning

A flagpole casts a shadow 18 feet long, while a 5-foot student standing nearby casts a shadow 3 feet long. Find the height of the flagpole.

$\frac{\text{height}}{18} = \frac{5}{3}$, so height $= \frac{18 \times 5}{3} = 30$ feet.

10 [Intermediate] • Target: Angle relationships

Two parallel lines are cut by a transversal. If one angle measures 72°, find its corresponding angle and its co-interior (same-side interior) angle.

The corresponding angle is equal: 72°. The co-interior angle is supplementary: $180^\circ - 72^\circ = 108^\circ$.

11 [Intermediate] • Target: Angle relationships

The three angles of a triangle are in the ratio $2 : 3 : 4$. Find each angle.

The parts sum to $2 + 3 + 4 = 9$, and the total is 180°, so each part is 20°. The angles are 40°, 60°, and 80°.

12 [Intermediate] • Target: Angle relationships

Two angles are complementary, and one is 34° more than the other. Find both angles.

$x + (x + 34) = 90$, so $2x = 56$, giving $x = 28$. The angles are 28° and 62°.

13 [Intermediate] • Target: Coordinate geometry: distance and midpoint

Find the distance between the points $(3, -2)$ and $(9, 6)$.

$d = \sqrt{(9-3)^2 + (6-(-2))^2} = \sqrt{36 + 64} = \sqrt{100} = 10$.

14 [Intermediate] • Target: Coordinate geometry: distance and midpoint

Find the midpoint of the segment connecting $(-5, 3)$ and $(7, -9)$.

Midpoint $= \left(\frac{-5+7}{2}, \frac{3+(-9)}{2}\right) = (1, -3)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 7.01 Trigonometry & Additional Topics

A circle has the radius shown below. Find its area and circumference, in terms of π.

Area $= \pi(8)^2 = 64\pi$. Circumference $= 2\pi(8) = 16\pi$.

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16 [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 8.01 Exponentials & Logarithms

Write the equation of a circle with center $(3, -4)$ and radius 6.

$(x-3)^2 + (y+4)^2 = 36$.

17 [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 1.01 Heart of Algebra

A circle has the radius and shaded sector shown below. Find the area of the shaded 60° sector, in terms of π.

Sector area $= \frac{60}{360} \times \pi(9)^2 = \frac{1}{6} \times 81\pi = 13.5\pi$.

18 [Challenging] • Target: Composite and shaded regions • Cross-Lesson: 2.01 Quadratics & Polynomials

A square with side length 12 has a circle of radius 5 removed from its center. Find the remaining area, in terms of π.

Square area $= 12^2 = 144$. Circle area $= \pi(5)^2 = 25\pi$. Remaining area $= 144 - 25\pi$.

19 [Challenging] • Target: Composite and shaded regions • Cross-Lesson: 3.01 Functions & Graphs

A rectangular garden 20 m by 15 m has a circular fountain of radius 3 m built inside it. Using $\pi \approx 3.14$, find the area of the garden not covered by the fountain, to the nearest whole number.

Garden area $= 300$. Fountain area $\approx \pi(9) \approx 28.26$. Remaining area $\approx 300 - 28.26 \approx 272$ square meters.

20 [Challenging] • Target: A well-designed word problem: designing a park • Cross-Lesson: 4.01 Ratios, Rates & Data

This question has three parts. A city is designing a park shaped like a rectangle with a semicircular section attached to one end. The rectangular section is 30 meters long and 20 meters wide, and the semicircle is attached to one of the 20-meter-wide ends. (a) Find the radius of the semicircle. (b) Find the total area enclosed by the park (rectangle plus semicircle), in terms of π. (c) The city fences the perimeter of the park: the three straight sides of the rectangle plus the curved semicircular edge, but not the shared 20-meter edge where the semicircle attaches. Find the total fence length needed, in terms of π.

(a) The semicircle's diameter equals the 20-meter width, so the radius is 10 meters. (b) Rectangle area $= 30 \times 20 = 600$. Semicircle area $= \frac{1}{2}\pi(10)^2 = 50\pi$. Total area $= 600 + 50\pi$ square meters. (c) The fenced sides are the two 30-meter lengths, the far 20-meter end, and the semicircular arc. The arc length is half the circumference: $\frac{1}{2}(2\pi \times 10) = 10\pi$. Total fence $= 30 + 30 + 20 + 10\pi = 80 + 10\pi$ meters.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Area and perimeter

A rectangle has length 23 and width 5. Find its area and perimeter.

a. $115, 56$ ✓

b. $56, 115$

c. $119, 56$

d. $115, 54$

Area $= 23 \times 5 = 115$ square units. Perimeter $= 2(23 + 5) = 56$ units.

22 [Foundational] • Target: The Pythagorean theorem

A right triangle has legs 3 and 4. Find the length of the hypotenuse c.

a. 5 ✓

b. 7

c. 3

d. 9

$c = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

23 [Foundational] • Target: Volume and surface area

In the coordinate plane, the circle $(x - 2)^2 + (y - 3)^2 = 25$ intersects the y-axis at two points. What is the distance between these two points?

a. $2\sqrt{21}$ ✓

b. $4\sqrt{5}$

c. 8

d. 10

For the y-axis, set $x = 0$: $(0 - 2)^2 + (y - 3)^2 = 25 \Rightarrow 4 + (y - 3)^2 = 25 \Rightarrow (y - 3)^2 = 21 \Rightarrow y = 3 \pm \sqrt{21}$. The distance between the points is $(3 + \sqrt{21}) - (3 - \sqrt{21}) = 2\sqrt{21}$.

24 [Foundational] • Target: Similar triangles and proportional reasoning

A sphere has a volume of 288π cubic centimeters. What is its surface area in square centimeters?

a. 144π ✓

b. 72π

c. 288π

d. 216π

Volume $\frac{4}{3}\pi r^3 = 288\pi \Rightarrow r^3 = 216 \Rightarrow r = 6$. Surface area = $4\pi r^2 = 4\pi(36) = 144\pi$.

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25 [Foundational] • Target: Angle relationships

What is the area of a circle whose circumference is equal to the perimeter of a square with side length 6?

a. $\frac{72}{\pi}$

b. $\frac{144}{\pi}$ ✓

c. 36π

d. 144π

Perimeter of square $= 4 \times 6 = 24$. Set circumference $2\pi r = 24 \Rightarrow r = \frac{12}{\pi}$. Area $= \pi r^2 = \pi\left(\frac{12}{\pi}\right)^2 = \frac{144}{\pi}$.

26 [Foundational] • Target: Coordinate geometry: distance and midpoint

In the coordinate plane, the circle $x^2 + y^2 = 25$ intersects the line $y = 3$ at two points. What is the distance between these two points?

a. 6

b. **8** ✓

c. 10

d. 12

Substitute $y = 3$: $x^2 + 9 = 25 \Rightarrow x^2 = 16 \Rightarrow x = \pm 4$. Points are $(-4, 3)$ and $(4, 3)$. Distance is $4 - (-4) = 8$.

27 [Foundational] • Target: Circles: area, circumference, and equation

A circle has circumference 10π. What is its area?

a. 5π

b. 10π

c. 25π ✓

d. 100π

$C = 2\pi r = 10\pi \Rightarrow r = 5$, $A = 25\pi$.

28 [Foundational] • Target: Composite and shaded regions

In a circle, a central angle of 60° intercepts what fraction of the entire circumference?

a. $\frac{1}{4}$

b. $\frac{1}{5}$

c. $\frac{1}{6}$ ✓

d. $\frac{1}{8}$

$\frac{60^\circ}{360^\circ} = \frac{1}{6}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: designing a park

The figure shows a circle with center O. The radius of the circle is 5 units. If the length of chord AB is 8 units, what is the distance from the center O to the chord AB?

a. **3 units** ✓

b. 4 units

c. 5 units

d. 6 units

Let O be the center of the circle and AB be the chord. Draw a perpendicular from O to AB, and let the point of intersection be M. This perpendicular bisects the chord AB. Thus, AM = MB = $\frac{8}{2} = 4$ units. We have a right-angled triangle OMA, where OA is the radius, OM is the distance from the center to the chord, and AM is half the length of the chord. By the Pythagorean theorem: $OA^2 = OM^2 + AM^2$ Given OA = 5 units and AM = 4 units: $5^2 = OM^2 + 4^2$ $25 = OM^2 + 16$ $OM^2 = 25 - 16$ $OM^2 = 9$.

30 [Intermediate] • Target: Area and perimeter

Triangle ABC is similar to triangle DEF, where each side of DEF is 4 times the length of the corresponding side of ABC. If the area of triangle ABC is 26 square units, what is the area of triangle DEF?

a. 104

b. 1664

c. 436

d. 416 ✓

For any two similar geometric figures, if the linear scale factor of side lengths is k, then the ratio of their areas is the square of the linear scale factor, k^2: $\frac{\text{Area}(DEF)}{\text{Area}(ABC)} = k^2 = 4^2 = 16$ Therefore:

$\text{Area}(DEF) = 16 \times \text{Area}(ABC) = 16 \times 26 = 416$ square units .

31 [Intermediate] • Target: The Pythagorean theorem

A circle has a radius of 18 cm. What is the area, in square centimeters, of a sector bounded by a central angle of 30°?

a. 27π ✓

b. 54π

c. 13π

d. 3π

The formula for the area of a sector with central angle θ (in degrees) and radius r is: $A = \frac{\theta}{360^\circ} \times \pi r^2$ Substitute $\theta = 30^\circ$ and $r = 18$: $A = \frac{30}{360} \times \pi(18)^2 = \frac{30}{360} \times 324\pi = 27\pi$ cm^2 .

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32 [Intermediate] • Target: Volume and surface area

A circle in the coordinate plane passes through $(0, 0)$, $(8, 0)$, and $(0, 6)$. What is the radius of the circle?

a. 4

b. **5** ✓

c. 6

d. 10

The triangle with vertices $(0, 6)$, $(0, 0)$, and $(8, 0)$ is a right triangle with right angle at the origin. By Thales's Theorem, the hypotenuse connecting $(0, 6)$ and $(8, 0)$ is a diameter of the circle. Diameter $d = \sqrt{8^2 + 6^2} = 10$. Thus radius $r = 5$.

33 [Intermediate] • Target: Similar triangles and proportional reasoning

A right circular cylinder has height 12 and base radius 5. A sphere is inscribed inside the cylinder such that it touches the cylinder's curved surface. What is the volume of the space inside the cylinder that is outside the sphere?

a. 100π

b. $\frac{400\pi}{3}$ ✓

c. $\frac{500\pi}{3}$

d. 200π

Sphere radius is $r = 5$. Cylinder volume $V_c = \pi r^2 h = \pi(25)(12) = 300\pi$. Sphere volume $V_s = \frac{4}{3}\pi(5^3) = \frac{500\pi}{3}$. Space outside sphere: $300\pi - \frac{500\pi}{3} = \frac{900\pi - 500\pi}{3} = \frac{400\pi}{3}$.

34 [Intermediate] • Target: Angle relationships

The line $x = 5$ intersects the circle $(x - 2)^2 + (y + 1)^2 = 25$ at two points. What is the distance between these two points?

a. 6

b. **8** ✓

c. 10

d. 12

Substitute $x = 5$: $(5-2)^2 + (y+1)^2 = 25 \Rightarrow 9 + (y+1)^2 = 25 \Rightarrow (y+1)^2 = 16 \Rightarrow y + 1 = \pm 4 \Rightarrow y = 3$ or $y = -5$. Distance $= 3 - (-5) = 8$.

35 [Intermediate] • Target: Coordinate geometry: distance and midpoint

What is the circumference of a circle with a diameter of 14 inches?

a. 7π in

b. 14π in ✓

c. 28π in

d. 49π in

Applying Area, Volume, Triangles, Circles & Trigonometry principles, $C = \pi d = 14\pi$ in. Therefore, the verified correct option is 14π in.

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36 [Intermediate] • Target: Circles: area, circumference, and equation

A chord in a circle of radius 10 cm is at a distance of 6 cm from the center. What is the length of the chord?

a. 8 cm

b. 12 cm

c. **16 cm** ✓

d. 18 cm

The perpendicular from the center bisects the chord. Half-length $= \sqrt{10^2 - 6^2} = 8$ cm. Full chord length $= 2 \times 8 = 16$ cm.

37 [Intermediate] • Target: Composite and shaded regions

An equilateral triangle has sides of length 10. What is its perimeter?

a. 20

b. 25

c. **30** ✓

d. 35

Applying Area, Volume, Triangles, Circles & Trigonometry principles, Perimeter = $3 \times 10 = 30$. Therefore, the verified correct option is 30.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: A well-designed word problem: designing a park • Cross-Lesson: 2.01 Quadratics & Polynomials

A circle with radius 8 has a sector with area 16π. What is the measure of the central angle of this sector in radians?

a. $\frac{\pi}{4}$

b. $\frac{\pi}{2}$ ✓

c. $\frac{2\pi}{3}$

d. π

Area $= \frac{1}{2}r^2\theta \Rightarrow 16\pi = \frac{1}{2}(64)\theta = 32\theta \Rightarrow \theta = \frac{16\pi}{32} = \frac{\pi}{2}$.

39 [Challenging] • Target: Area and perimeter • Cross-Lesson: 3.01 Functions & Graphs

A circle has equation $(x-2)^2 + (y+1)^2 = 16$. What is the radius?

a. 2

b. **4** ✓

c. 8

d. 16

Applying Area, Volume, Triangles, Circles & Trigonometry principles, $r^2 = 16 \Rightarrow r = 4$ Therefore, the verified correct option is 4.

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40 [Challenging] • Target: The Pythagorean theorem • Cross-Lesson: 4.01 Ratios, Rates & Data

A circle has radius 12 inches. What is the area of a sector with a central angle of 60°?

a. 12π in^2

b. 18π in^2

c. 24π in^2 ✓

d. 36π in^2

Sector Area $= \frac{60^\circ}{360^\circ} \times \pi(12^2) = \frac{1}{6} \times 144\pi = 24\pi$ in^2.

41 [Challenging] • Target: Volume and surface area • Cross-Lesson: 5.01 Statistics & Probability

A sector of a circle with radius 12 cm has an area of 48π square centimeters. What is the central angle of the sector in degrees?

a. 120° ✓

b. 60°

c. 90°

d. 150°

Total area = 144π. Ratio = $48\pi/144\pi = 1/3$. Angle = $\frac{1}{3} \times 360^\circ = 120^\circ$.

42 [Challenging] • Target: Similar triangles and proportional reasoning • Cross-Lesson: 7.01 Trigonometry & Additional Topics

In a $30^\circ - 60^\circ - 90^\circ$ triangle, the side opposite the 30° angle is 7. What is the length of the side opposite the 60° angle?

a. $7\sqrt{2}$

b. $7\sqrt{3}$ ✓

c. 14

d. $14\sqrt{3}$

In a 30-60-90 triangle, sides are in ratio $1 : \sqrt{3} : 2$. Side opposite 60° is $7\sqrt{3}$.

43 [Challenging] • Target: Angle relationships • Cross-Lesson: 8.01 Exponentials & Logarithms

A central angle of 45° in a circle of radius 16 cm intercepts an arc. What is the length of this arc?

a. 2π cm

b. 4π cm ✓

c. 6π cm

d. 8π cm

$s = \frac{45^\circ}{360^\circ} \times 2\pi r = \frac{1}{8} \times 32\pi = 4\pi$ cm.

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44 [Challenging] • Target: Coordinate geometry: distance and midpoint • Cross-Lesson: 1.01 Heart of Algebra

A cylinder has a radius of 4 cm and a height of 9 cm. What is its total surface area in terms of π?

a. 104π ✓

b. 72π

c. 144π

d. 96π

Total surface area = $2\pi rh + 2\pi r^2 = 2\pi(4)(9) + 2\pi(16) = 72\pi + 32\pi = 104\pi$.

45 [Challenging] • Target: Circles: area, circumference, and equation • Cross-Lesson: 2.01 Quadratics & Polynomials

What is the surface area of a cube with edge length 4 cm?

a. 64 cm²

b. **96 cm²** ✓

c. 128 cm²

d. 144 cm²

A cube has 6 congruent square faces: $SA = 6s^2 = 6(4^2) = 6(16) = 96 \text{ cm}^2$.

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